Supplementary Material

Cognitive Edge Computing

A Survey on Optimizing Large Language Models and Autonomous Agents for the Edge

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This supplement separates recovered measurements (S1), proposed deployment/reporting requirements (S2–S4, S6, S10, S12), and inspectable source/artifact records (S5, S7–S9, S11). Analytical and conceptual figures are labelled accordingly; none supplies an unmeasured performance result. References are local to this supplement.

The public companion edition provides the rendered supplement, figure inputs, released empirical records and executable checks listed in S7. Editable submission sources and author-local build files are not distributed. S9 follows the seventeen records in main Tables IV–V; its grading ledger states the result and inspected replication support. S10 groups additional reporting fields by execution mechanism.

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S1 Exploratory Historical Server Measurements

S1.1 Role of the Server Record and Recovered Provenance

The previously submitted 4×RTX 4090 D results are retained, not replaced by synthetic examples. A renewed archive inspection recovered thirteen server-suite CSV/summary pairs, their source-code task definitions, the runner, and the historical selection script. These contain 2350 logged requests and 189 recorded errors across separate campaigns. The complete five-model campaign contains 1200 requests: five models, four context ceilings, ten fixed tasks, and six repeated requests per model–context–task cell. A separate 200-request Qwen3 8B follow-up has five repeat indices per cell. The campaigns are not pooled.

The recovered logs support an **E2 descriptive reanalysis**: their recorded scores and timing summaries can be recalculated, but exact checkpoint digests, per-model precision, runtime/driver versions, placement, command-line settings, and complete streaming/power traces remain unavailable. The original matched-capability and edge-performance interpretations remain unsupported E3 claims and are withdrawn. Neither these historical results nor their reanalysis derives the survey's deployment recommendations. No model or edge device was newly executed.

S1.2 Preserved Submission-Era Summary and Campaign Mixing

Table S1 preserves the submitted values. Four rows originate from the complete five-model campaign; the Qwen row instead reflects the later single-model follow-up selected by the archived latest-complete-per-model procedure. Old Qwen plots used the earlier campaign, explaining their approximately 1301 ms context-averaged p95 versus the submitted 2380.3 ms. This resolves the provenance discrepancy without making the two executions equivalent. The original "Qual." label denotes normalized answer containment, not strict exact match. Context-averaged p95 is not a pooled-request percentile. Figure S1 recomputes each campaign separately from its requests.

Table S1: Preserved submission-era summaries. These combine different campaigns; they are retained for provenance, not as a matched comparison. Mean p95 averages context-specific quantiles. J/tok is an endpoint-power proxy.

Model	Qual.	Tok/s	Mean p95 (ms)	Approx. J/tok
DeepSeek-R1 14B	0.150	94.0	1885.9	5.70
Gemma3 4B	0.917	224.6	634.6	12.26
GPT-OSS 20B	0.983	165.0	1023.0	3.50
Llama2 7B	0.650	234.5	534.0	2.87
Qwen3 8B	0.220	137.1	2380.3	5.75

S1.3 Recovered Configuration and Remaining Gaps

Table S2 replaces blanket missing-record statements with the information actually recovered. Defaults in a surviving script are not evidence of the options passed during a historical execution.

Table S2: Recovered server provenance and unresolved execution metadata.

Item	Recovered record	Remaining limit
Platform	Reported four-board RTX 4090 D server, 24 GB per board; Ollama API runner.	No board identifiers, placement, concurrency trace, runtime/driver pin, or evidence of a matched mobile configuration.
Models	Five recorded tags: DeepSeek-R1 14B, Gemma3 4B, GPT-OSS 20B, Llama2 7B, Qwen3 8B.	Exact checkpoint/template digests and per-model weight/activation/KV precision are unrecorded.
Requests	1200 requests in the main campaign; 200 in the Qwen follow-up. Raw predictions, reference answers, latency, token counts, repeat indices, and errors survive.	Stored predictions flatten newlines; full HTTP responses, separate reasoning channels, termination reasons, and process interruptions are not preserved.
Workload	Ten task IDs and source-code prompts (Table S3). Context ceilings 512/1024/2048/4096. Main recorded input counts: 16–95 tokens.	Exact historical payload/template hashes are absent. A context ceiling is not an actual long-context workload.
Repetition	Six repeated requests per main cell; five per follow-up cell. The source uses a sequential task/repeat loop with an unlogged warm-up.	Repeat IDs do not establish independently restarted sessions; initial state, thermal state and warm-up costs are unknown.
Decoding	Observed output maxima: 128 tokens in the main campaign, 256 in the follow-up; Qwen reaches those counts in 240/240 and 160/200 rows.	These counts suggest a budget boundary; exact generation caps, thinking mode, sampling settings and stop reasons are not logged.
Timing	Wall-request time and decode rate survive. Archived `ttft_ms` comes from Ollama's prompt-evaluation duration.	That field is not measured end-to-end TTFT; no timestamped streaming trace exists.
Scoring	Source normalizes whitespace/case and tests whether the reference answer occurs as a substring. Every archived score is replayable.	Containment permits incidental matches. Strict string equality is a separately labelled post-hoc formatting audit, not semantic adjudication.
Power	The runner sums visible GPU power before and after a request; their arithmetic mean times wall time forms `energy_j`.	Two endpoints are not a power trace; board identities, sampler/server alignment, activity attribution and host/radio energy are absent.

S1.4 Ten-Task Suite and Scoring Definition

Table S3 transcribes the surviving source-code roster. All matching task IDs and answers appear in the two reported campaigns; the exact request payload is not independently archived. For a stored prediction p and reference a, the historical hit is $h = \mathbb{1}[n(a) \text{ is a substring of } n(p)]$, where n lowercases and collapses

whitespace. The post-hoc strict-string audit instead uses $\mathbb{1}[n(a) = n(p)]$. Neither metric alone verifies reasoning or an agent's action.

Table S3: Recovered source-code task definitions and reference answers.

Task	Prompt in the recovered runner	Answer
`arith_add`	Compute 27 + 15. Return only the integer.	42
`arith_div`	Compute 144 / 12. Return only the integer.	12
`logic_set`	All roses are flowers. Some flowers fade quickly. Can we conclude all roses fade quickly? Answer Yes or No only.	No
`count_chars`	How many letters are in the word EDGE? Return only the integer.	4
`calendar`	If tomorrow is Wednesday, what day is today? Return only the day name.	Tuesday
`step_cost`	A pen costs 3 dollars. You buy 4 pens and use a 2 dollar coupon. Final cost? Return only the integer.	10
`reverse`	Reverse this string: abcde. Return only the reversed string.	edcba
`compare`	Which is larger, 0.125 or 0.5? Return only one value exactly as written.	0.5
`short_reason`	A train travels 60 km in 1.5 hours. What is the average speed in km/h? Return only the integer.	40
`boolean`	Is 11 a prime number? Answer Yes or No only.	Yes

S1.5 Retrospective Results and Descriptive Variability

Figure S1 preserves the empirical information with a consistent request-level reanalysis. For each main-campaign model, $n = 240$ (ten tasks, six repeated requests, four context ceilings); the follow-up has $n = 200$. Each of these two campaigns records no transport error, but an error-free response can still be empty or wrong. All 1400 rows enter their campaign-specific summaries. Median and linearly interpolated p95 use individual wall times, not averages of context-specific percentiles. Decode-rate SD describes heterogeneity across recorded requests, not uncertainty in a population mean. The published CSV also reports wall-time request SD and SD of repeat-index mean wall times: respectively 100.4/32.0, 75.3/24.9, 126.6/19.4, 150.7/16.0, and 98.6/21.2 ms for the five main-campaign models in table order, and 99.9/13.9 ms for the follow-up. The same CSVs additionally give sample SD across repeat-index aggregates of containment, strict-string score, mean decode rate and endpoint-proxy J/token (sum of proxy joules divided by summed output tokens within each index). Containment SD ranges from 0 to 0.084 across the main models and is 0.045 in the follow-up. A zero SD describes identical aggregate scores on these fixed tasks, not certainty about general capability. Energy-proxy variation does not estimate calibration bias or missing within-request power. These indices are technical repetitions, not independent deployment campaigns; no population confidence interval or significance test is inferred.

The large score gaps have observable measurement correlates. The main Qwen run stores 240 empty response fields with 128 decoded tokens each; the follow-up stores 156 empty fields and 44 exact-answer fields, producing 0.22 containment and strict-string scores. DeepSeek stores 132 empty fields and frequently explanatory or partial text; its containment score is 0.15 but strict-string equality is zero. Llama's 0.65

containment versus 0.35 strict-string rate exposes formatting sensitivity; Gemma's corresponding rates are 0.917 and 0.80. These facts do not identify whether a missing final answer arose from a token limit, a separate unrecorded reasoning channel, template behavior, or another execution setting. No causal explanation or ranking of intrinsic reasoning ability follows.

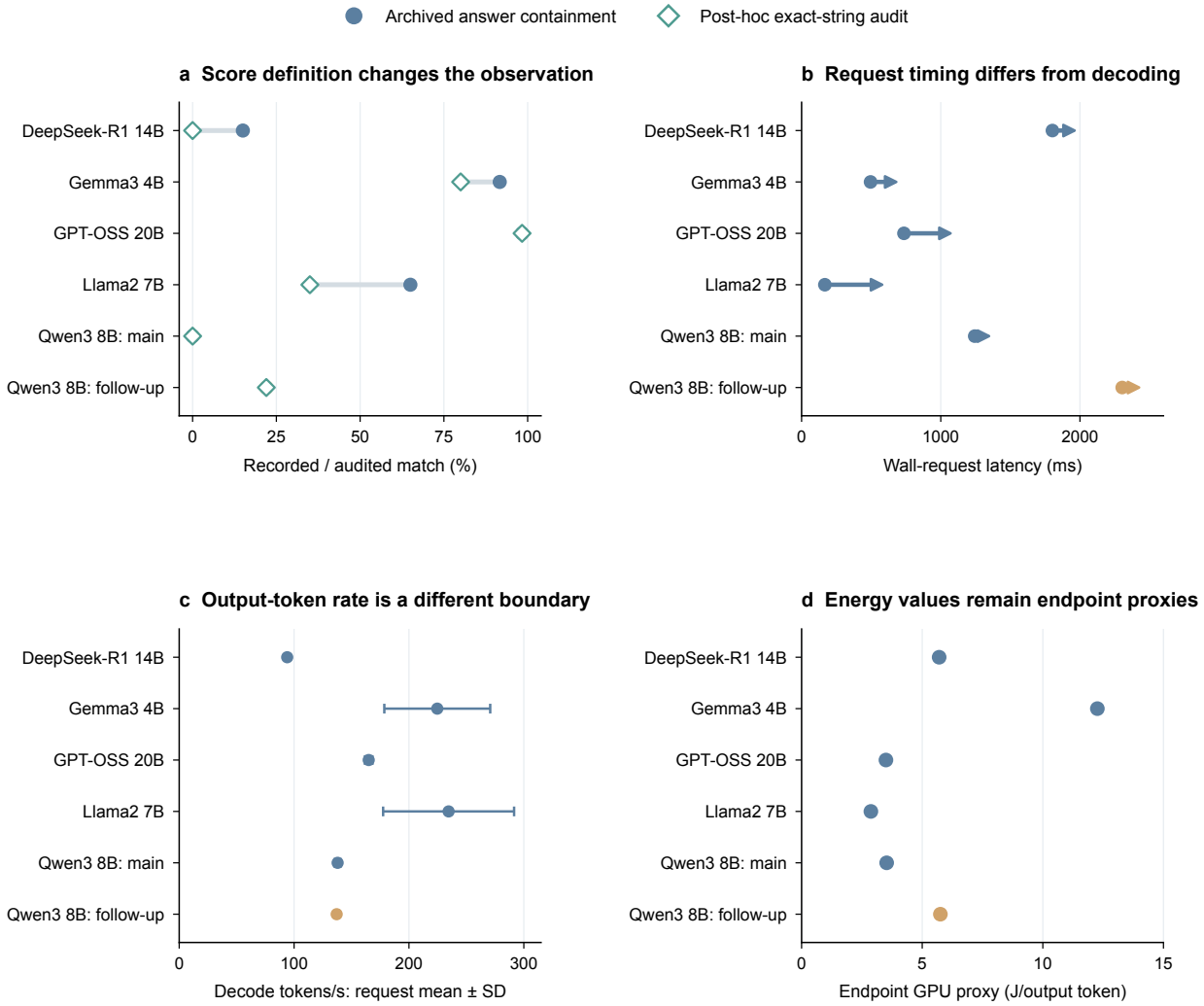


Figure S1: Recovered server-request results, separated by campaign. Blue rows use the 1200-request main campaign (240/model); the amber Qwen row uses the separate 200-request follow-up. (a) Archived answer containment versus a post-hoc strict-string formatting audit. (b) Request median (circle) to linear-interpolated pooled p95 (triangle), not confidence intervals. (c) Mean decode rate with one request SD; requests span fixed tasks and repeat indices. (d) Sum of endpoint-power energy proxies divided by summed output tokens, without a validated energy uncertainty estimate. These are log-level descriptions of recorded executions, not matched model rankings or mobile measurements.

S1.6 Context, Timing, and Endpoint-Power Boundaries

Figure S2 explains three limits that change interpretation. Actual main-campaign prompt counts are only 16–95 despite ceilings of 512–4096, so these are configuration sweeps over short prompts, not long-context retention tests. The runner uses a non-streaming call; its `ttft_ms` field is prompt-evaluation duration and must not be labelled observed user TTFT. Its GPU energy proxy is

$$\widetilde{E}_i = \frac{P_{i,\text{before}} + P_{i,\text{after}}}{2}\, T_{i,\text{wall}}, \qquad \widetilde{e}_{\text{tok}} = \frac{\sum_i \widetilde{E}_i}{\sum_i N_{i,\text{out}}}. \tag{1}$$

The code sums power over boards visible to `nvidia-smi` on the runner host. Only their endpoint mean survives, and the logs do not establish board IDs or sampler/server co-location. This proxy cannot reconstruct within-request transients, attribute other activity, integrate idle/cooling/host/radio cost, or establish system energy. It is also not the ratio of context-averaged board power to decode-only token rate. The reported J/token values remain labelled unvalidated proxies.

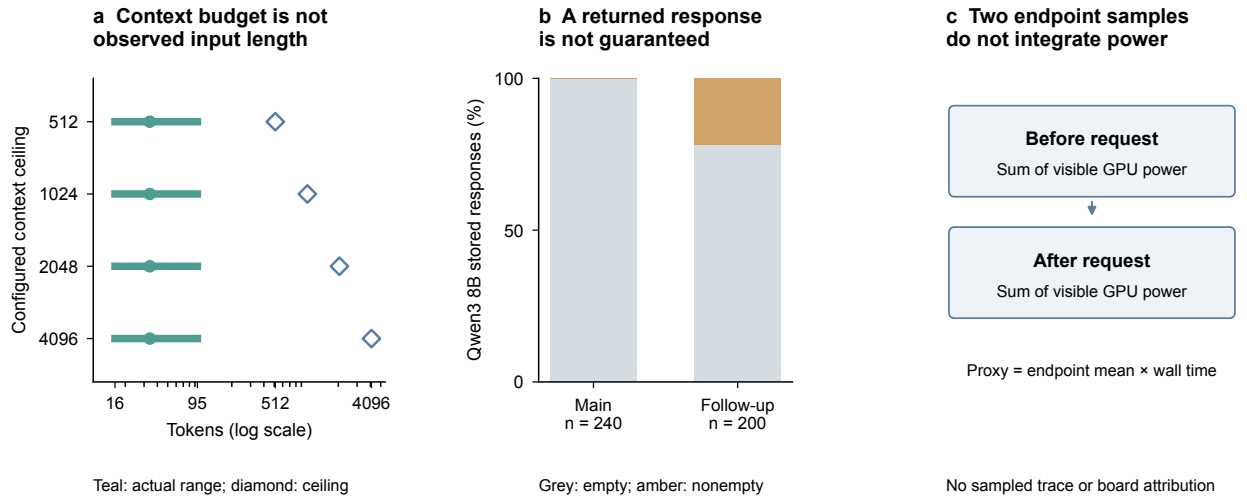


Figure S2: Interpretation checks from the recovered record. (a) Actual prompt-token min–max and median versus configured context ceiling, using all 1200 main-campaign rows. (b) Empty versus nonempty stored Qwen response fields in the main and follow-up campaigns; nonempty is not automatically correct. (c) The source-code endpoint-power procedure, whose missing within-request trace and device attribution prevent a validated energy inference. Geometry in (c) is conceptual; the other panels are derived from the archived requests.

S1.7 Failure Ledger, Reproduction, and Remaining Interpretation

The public campaign ledger retains all thirteen recovered server-suite files, including connectivity probes and partial/pilot campaigns. Its 2350 rows include 189 recorded errors; these belong to different campaigns and are not pooled into a system failure probability. The earlier selector excluded model–context summaries with a recorded failure and selected a latest complete campaign per model. The reanalysis instead preserves each file's denominator and labels its role. Warm-ups are omitted by the historical logger, and process interruptions or unarchived attempts cannot be counted; the ledger is therefore a census of recovered rows, not proof of complete operational logging.

The server archive guide links the thirteen CSV/summary pairs, hash/normalization inventory, task roster, runner, selection source, derived results, and exact reanalysis commands. The scorer and saved successful-summary percentiles are checked against every recovered campaign. Request-log recalculation is reproducible; re-executing an identical model/device experiment is not, because execution metadata remains incomplete. The separate synthetic format fixture in S7 is solely a helper example and supplies no historical observation. For cognitive edge computing, the lesson is to declare task acceptance, actual input/output work, observable response channels, complete failure accounting, and power boundaries before transferring a score or token rate to a resource-constrained agent.

S2 Extended Application Scenarios

The following component-grounded scenarios extend the main application table. The cited sources validate their own components and settings; the complete applications below are prospective compositions, requiring matched workflow tests.

Private text and action assistant.
A compact text model can interpret a request, while a specialized function caller generates a bounded API action. Qwen3 and FunctionGemma provide concrete components [1, 2, 3]. Source-fact correctness, permission, confirmation and a unique action receipt define completion; S12 specifies this proposed contract.

Local visual assistant.
FastVLM supplies an encoder/token co-design and device-oriented export path [4]. A camera assistant must additionally validate the correct image fact, image resolution, sensor-to-answer latency and sustained resident state. The paper's VLM benchmark is not validation of an unrestricted embodied agent.

Network cache controller.
Cooperative caching uses an LLM to propose bounded cache changes under a parser/fallback interface [5]. Deployment must separately measure decision freshness, traffic utility and controller deadline/energy. The reported simulated network utility does not establish those device costs.

S3 Expanded Practitioner Checklist Rationale

W1–W4 rationale (expanded from the main text).

- **W1** (joint reporting): report task quality and attempted-task success together with deadline satisfaction, request and task latency, and task energy under a declared boundary. Mean energy per attempt and amortized energy per success include failed attempts (definitions in S7). Token efficiency is useful only alongside task-level outcomes; it cannot establish successful-task efficiency by itself. Energy-scope disclosure follows the measurement motivation in [6]; the episode denominators here are explicit accounting definitions, not new observations.
- **W2** (long-context robustness): test distant-fact access separately from stable recent-window generation. StreamingLLM and InfLLM address different state-retention mechanisms [7, 8]; evaluate critical facts under the actual eviction/retrieval policy.
- **W3** (agent-loop safety): record tool correctness, recovery and unauthorized actions under an explicit permission/threat model. InjecAgent and AgentDojo provide concrete attack/utility evaluation settings [9, 10]; the proposed edge test must also account for retries and defense cost.
- **W4** (offload governance): disclose placement, link conditions and state ownership; EdgeShard demonstrates setting-dependent partition behavior [11]. Data residency, jitter and thermal degradation are additional deployment checks, not effects already established by that prototype.

S4 Deployment Characteristics Across Computing Tiers

This proposed accounting table extends the main placement discussion. Edge computing distinguishes local, nearby and remote execution [12]; energy reporting must state its measured scope [6]. The cells specify

what a new study should report for each role. It is a qualitative accounting framework, not a measurement from the historical server record or a ranking of devices.

Table S4: Proposed reporting boundaries across execution tiers. Tiers denote execution roles, not parameter-count classes or measured performance ranks.

Aspect	Remote cloud service	Nearby edge server	Client device
Resource control	Shared/elastic compute; provider admission and queueing	Locally managed shared compute; contention and capacity limits	Fixed available memory and sustained thermal/power envelope
Network path	WAN transfer and remote service availability	Access-link and local-network transfer	Local inference may operate offline; delegation remains conditional
Memory accounting	Model shards, KV/session load, service overhead	Resident models, concurrent KV and runtime buffers	Weights, KV, application/OS reservation, sensor/tool state
Latency reporting	End-to-end TTFT, decode and queueing under the measured WAN	Same phases under the measured link/session arrival process	Prefill/decode and complete-task latency under thermal load
Energy boundary	Name measured server components; keep facility overhead separate	Identify server, networking and host scope	Identify SoC/device/radio scope and task completion denominator
Trust boundary	State which data/tool states leave the client and provider permissions	State operator ownership, isolation and local residency	Locality alone does not establish confidentiality or action safety

S5 Survey Scope, Retained-Link Audit, and Reference Inventory

The protocol summary and retained-corpus audit are published as `survey/protocol.md` in the official repository https://github.com/wangxb96/Awesome-EdgeAI. The dated census uses the archived manifests at immutable commit `82b6110`, rather than the changing default branch. It states source classes, query-concept families and an example composite query, inclusion/exclusion criteria, E1/E2/E3 grading, and the limits of the available screening record summarized in Sec. “Survey Methodology and Evidence Grading” of the main manuscript.

Table S5: Retained-corpus census (snapshot 11 November 2026), anchored to archived manifests at commit `82b6110`.

Census item	Count
Unique arXiv identifiers in the frozen census	145
Resolved against the arXiv Atom API	145 / 145
Unresolved identifiers	0
Withdrawn preprints excluded	0
Markdown link labels title-audited	137
Distinct recorded GitHub URL-root strings	123
Recorded URL strings resolving to a live Git HEAD	123 / 123
Audited arXiv corpus items from 2023–2026	123 / 145 (84.8%)
Repository roots after case folding (separate recount)	122
References cited in main / supplement (13 November)	96 / 30
Records in broader BibTeX file (13 November)	275

The 145 unique arXiv records, 137 title-audited Markdown labels, and 123 distinct recorded repository URL-root strings are different counting units, not successive screening totals. Case folding collapses the two spellings of `google-ai-edge/LiteRT-LM`, giving 122 normalized roots; it does not resolve repository renames or aliases. The frozen manifest remains unchanged. A live Git HEAD verifies repository reachability, not official ownership or replication sufficiency. The cited-reference index lists the 96 main references and their numbering; the supplement cites 30 of these same entries. These are references, including surveys and documentation, not 96 independent experimental studies. The broader 275-entry BibTeX file and seventeen purposive matrix records are different collections. Evidence grades combine publication maturity and artifact support. E1 requires archival publication with public repeatable artifacts; E2 denotes inspectable quantitative evidence with partial replication, irrespective of publication status; E3 denotes exploratory or incomplete evidence. The protocol is a reproducible audit of retained, primary-source-verifiable items rather than a PRISMA review. Historical per-database query logs, exact source-specific dates, retrieval totals, and screening/exclusion counts were not retained; the published scripts cannot reconstruct them. This limitation is stated in the manuscript's Threats to Validity section.

S6 Operationalized Cognitive Capabilities

The C1–C5 identifiers and names match main Table III. These are proposed acceptance and diagnostic fields, not five experimentally validated cognitive scores. Quality/resource claims must use a matched task and device population. Security is a cross-cutting threat condition rather than a replacement for C5; attack benchmarks establish particular evaluation settings [9, 10].

Table S6: Capability acceptance criteria, diagnostic proxies and proposed edge stressors.

Capability	Task-level acceptance	Diagnostics and stressors
C1 Reasoning/answer fidelity	Correct task answer under the declared scorer; timeliness reported separately.	Next-token accuracy, perplexity and numerical equivalence are proxies, not complete reasoning tests. Vary precision and output budget.
C2 Context/task memory	Use the required earlier fact correctly, with source/freshness checks.	Retrieval hits and streaming perplexity diagnose different mechanisms. Vary fact distance, KV eviction and stale/missing evidence.
C3 Planning/tool completion	Reach the goal with valid tool arguments and verified terminal state.	Call correctness, retries and rollback are diagnostics. Vary tool timeout, network outage and available action schemas.
C4 Controlled autonomy	Complete only permitted actions with required confirmation; report unsafe/duplicate actions.	Intervention and refusal rates require utility context. Vary permissions, malicious tool content and ambiguous action receipts.
C5 Robustness/adaptation	Retain declared task success after specified distribution/resource changes or updates.	Measure task regressions, recovery and update overhead under memory pressure, thermal load and changing inputs; no universal robustness score is assumed.

S7 Artifact Reproduction Instructions

The public companion includes the released server records, analysis code, figures, protocols and final supplement PDF listed below. Editable submission sources and author-local TeX build files are excluded. The artifact edition is pinned to tag `public-artifacts-2026-11-13-v5`; use a disposable checkout for generated analysis outputs. This edition, the frozen census at `82b6110`, and the unchanged request helper at `24d5442` have different roles. Third-party studies and model artifacts remain linked to their original publishers.

S7.1 Public File Index

Resource	Published location and purpose
Supplement PDF	Rendered supplement; editable submission sources and local TeX template/build inputs are not part of the public repository.
Protocol and bibliography	Survey protocol, benchmark contract, and bibliographic database. The database is broader than the main paper's printed reference list.
Manifests and link audits	Frozen literature manifest, frozen repository manifest, literature audit, and repository audit; living-guide results use separate current_*.

Resource	Published location and purpose
Figure sources and data	Figure catalogue links every drawing script and export, including QSpec replot code, its reported score CSV, and the analytical memory CSV.
Historical server archive	Archive guide, hash inventory, reanalysis code, and all-campaign ledger; these are recovered observations, distinct from the synthetic fixture below.
Runnable format example	Request helper, synthetic input, and expected descriptive output. These fabricated format examples are not measurements or evidence of model performance.
Environment and checks	Figure dependencies, public-file inventory, and file-closure check. Python-only audits use the standard library; the distributed supplement is a rendered PDF and has no public TeX build inputs.

S7.2 Reproduce the Published Artifact Edition

```
git clone https://github.com/wangxb96/Awesome-EdgeAI.git
cd Awesome-EdgeAI
git checkout --detach public-artifacts-2026-11-13-v5
python3 survey/check_docs.py
python3 survey/check_public_artifacts.py
python3 survey/server/analyze_archived.py
python3 survey/compute_ci.py \
  --input survey/examples/request_records_synthetic.csv \
  --out request_summary_local.json
```

The generated `request_summary_local.json` must match the published expected output; it summarizes a synthetic formatting fixture only. To redraw the figures, install the pinned packages with `python3 -m pip install -r survey/requirements-figures.txt` and use the catalogue's commands. No checkpoint or device benchmark is executed by these checks. An actual study must supply its own complete request log, task roster, prompts, scorer, traces, configuration, and hashes under the benchmark contract. The request helper summarizes supplied fields; it does not integrate power, validate scoring, discover unlogged attempts, infer complete-task success, or infer missing S1 execution metadata. Its legacy JSON keys containing `task` count constituent request attempts, as defined in the benchmark contract.

Confidence intervals are off by default. Only when experimental design justifies independent repetition IDs should a new measured run use `-bootstrap-unit repetition_id -bootstrap 1000`. This estimates an equally weighted mean of repetition-specific summaries; it is neither a pooled-request percentile interval nor validation of independence. The synthetic fixture supplies no such experimental repetitions.

S7.3 Reproduce the Separate Frozen Census

```
# In a separate disposable checkout:
git checkout --detach 82b6110
python3 survey/verify_corpus.py --offline
python3 survey/verify_repos.py --offline
```

These commands replay archived metadata and repository status for the dated census, not historical database searches or today's network state. Running the audits without `-offline` produces a new live audit that must be dated and pinned separately. The current request helper is absent from this older snapshot; return

to the published artifact edition for request analysis. The recovered S1 logs have a separate analysis; absent execution metadata remains unrecorded.

S7.4 Task Episodes, Requests, and Energy Denominators

The main manuscript's S/A concerns complete task episodes. A task episode includes internal model/tool calls, fallback, and retries until its declared terminal outcome; a new user-level restart is a new episode. Keep an episode ledger with success criteria, task start/termination timestamps, deadline, and energy boundary, and link each constituent request through `episode_id` and `parent_request_id`. Those identifiers are optional metadata for the request helper, which does not aggregate workflows or infer goal completion. Do not mix episode and request rows in one helper analysis. For task deadline satisfaction, unsuccessful or aborted episodes do not count as successful completion at their finite termination time; retain their actual termination times and costs separately. Shared-device energy requires an explicit allocation rule to avoid counting it twice across overlapping episodes. For $A > 0$ attempted episodes, S verified successes and measured episode energy E_i, report mean attempt energy $\sum_{i=1}^{A} E_i/A$ and amortized energy per success $\sum_{i=1}^{A} E_i/S$ ($S > 0$). Both numerators include failed episodes and internal retries. At $S = 0$, the per-success ratio is undefined. If reporting $\sum_{i \in \mathcal{S}} E_i/S$ for the successful subset $\mathcal{S}$, name it the successful-episode conditional mean; it excludes failure energy and answers a different question.

S8 Source Versions for the Survey-Scope Comparison

The main manuscript's table distinguishes seven independent related surveys from the initial preprint version of the present manuscript. It describes inspected scope rather than assigning binary coverage scores. The following source versions fix the applicability of its section anchors. U means that a feature was not established from the inspected material, not that the work lacks it. Resource catalogues, executable implementations, and frozen corpus audits are distinct artifact types. Scope anchors were checked on 11 November 2026; formal publication metadata were updated on 13 November. Author-version section anchors below are kept separate from archival pagination.

Wang et al., Empowering Edge Intelligence (authors' prior survey). Author fulltext https://arxiv.org/html/2503.06027v2 (17 March 2025); archival DOI 10.1145/3724420. Anchors refer to the author version: smartphone/IoT applications (3.1–3.2), memory/energy (4.2–4.3), privacy/security (4.5, 6.4), and model/system optimization (5.2–5.3).

Initial preprint version of this manuscript, Wang and Jia, Optimizing Edge AI. https://arxiv.org/html/2501.03265v1 (4 January 2025). This is the first preprint version of the present manuscript, not an independent prior survey. It is shown solely to document version provenance and is excluded from the count of seven independent related surveys. The comparison deliberately inspects v1. Anchors are the data–model–system triad (III–V), resource collection (I-B), hardware (V-B), resource limits (VII-A–C), and security/privacy (VII-E).

Zheng et al., A Review on Edge Large Language Models. Author fulltext https://arxiv.org/html/2410.11845v2 (23 February 2025); archival DOI 10.1145/3719664. Author-version anchors cover comparative selection (3.6), runtime scheduling/frameworks (4.1), CPU/GPU/NPU hardware (4.3), personal agents (5.1.1), and multi-agent directions (6).

Xu et al., On-Device Language Models. Author fulltext https://arxiv.org/html/2409.00088v2 (14 September 2024). Anchors cover performance indicators (2.5), architectures and collaboration (3, 3.3), compression (4), hardware and frameworks (5), function-calling examples (6.1), and data security (7.1). The introduction links the authors' resource collection.

Yang et al., Decision-Making Large Language Model for Wireless Communication. DOI 10.1109/COMST.2025.3614494. *Primary-abstract-only boundary*: the IEEE abstract establishes data construction, architecture adaptation, reasoning/inference, decision frameworks, multi-agent coordination, and edge adaptability. Fulltext/TOC access was unavailable; no section-number or detailed hardware/evaluation/security claim is inferred. The coauthor scope statement at https://botwwt.github.io/publications/ corroborates the broad scope only.

Yang et al., Beyond the Edge. Author fulltext https://arxiv.org/html/2404.14238v1 (22 April 2024); archival DOI 10.1109/COMST.2024.3405075. Author-version anchors concern single/multi-agent RL (II), network requirements (III), offloading/caching/communication (IV), platforms (VI-A), and security/privacy (III-F, VI-G). These agents are RL policies, not implicitly LLM tool-use agents.

Cai et al., Tutorial on LLM-Enhanced Reinforcement Learning. Author-version anchors: https://arxiv.org/html/2512.03722v1 (3 December 2025). The archival record is COMST 28, 6665–6693 (2026), DOI 10.1109/COMST.2026.3712862; its published 11-author list is used in the bibliography. Anchors cover LLM roles (III), network case studies (IV–VI), edge-deployable architectures (VII-B), trust/security (VII-C), and collaboration (VII-D). Section III, footnote 6, links a dataset/code resource list.

Zhang et al., Toward Edge General Intelligence with Agentic AI and Agentification. Author-version anchors: https://arxiv.org/html/2508.18725v1 (26 August 2025). The archival record is COMST 28, 4285–4318 (2026), DOI 10.1109/COMST.2026.3651702. Anchors cover budget-aware design principles/checklists (I-B), agent framework (II), compactness, energy, connectivity and reasoning (III), open-source projects (IV), numerical case studies (V), and federated privacy directions (VI).

The table acknowledges substantial overlap. This survey's specific added contribution is the primary-study-to-conditional-rule evidence synthesis under deployment budgets and a frozen retained-corpus audit. Neither broad topic coverage nor a universally optimal optimization stack is claimed as new.

S9 Primary-Study and Plotting-Source Traceability

Records below follow main Tables IV–V. Selection is purposive: the seventeen entries illustrate distinct optimization controls, execution tiers and evidence types with inspectable results. They are representative synthesis examples, not all eligible literature or a reconstructed screening flow. Artifact inspection is dated 13 November 2026; no third-party result was independently rerun. Publication type and result-specific artifact support are stated explicitly; grades combine these requirements. Mobile transfer is assessed separately.

S9.1 Result-Specific Grading Rule and Inspection Ledger

E1 requires an archival quantitative result and a public implementation plus identifiable evaluation/benchmark path supporting a rerun in principle. E2 applies when that result's path remains incomplete or unverified, even for an archival paper. A row that invokes several outcomes inherits the least established replication path among them. Generic benchmark availability does not establish correspondence to a reported workload or comparator. Lack of a phone experiment is a transfer limit and does not itself lower the grade. "Unverified" means this inspection did not establish a component; it does not assert that the component does not exist. Current code availability does not pin the historical experiment. The public result-specific ledger records the result, source version, inspected components, missing/unverified links and revision/date for every entry.

S9.2 Model, Quantization, and Kernel Records (Table IV)

AWQ, MLSys 2024, Secs. 4–5 and Figs. 9–10. Quantization/evaluation and TinyChat: https://github.com/mit-han-lab/llm-awq. [13].

MobileQuant (E2), EMNLP Findings 2024, Sec. 5.6 and Tables 5–6. The W8A8 result retains higher precision for sensitive nonlinearities, rather than quantizing every operator to 8 bits. The LAMBADA proxy uses the first 1000 last-token prediction samples; costs are single forward passes under sustained execution. The paper describes sustained battery-discharge measurement, but the public acquisition/aggregation path behind the quoted energy was not established; the combined row is E2. Task/device profiling code: https://github.com/saic-fi/MobileQuant. [14].

FlashAttention-2 (E2), ICLR 2024, Sec. 4. Numerical tests and FA2 kernel timing are public, but a matched FA1 comparator for the quoted speedup was not established; the combined row is E2. Source: https://github.com/Dao-AILab/flash-attention. [15].

PagedAttention/vLLM (E2), SOSP 2023, Secs. 6–7 and Table 1. Serving code: https://github.com/vllm-project/vllm. Generic serving benchmarks, including the inspected v0.2.0 paths, do not establish the cited comparison's original workload/baseline linkage. [16].

PowerInfer-2, https://arxiv.org/html/2406.06282v2, Sec. 7. Author project: https://powerinfer.ai/v2/. Availability of a matching smartphone runtime was not verified; desktop PowerInfer code is not treated as that artifact. [17].

FastVLM (E2), CVPR 2025, Fig. 1 and experimental settings. Inference/checkpoints/export are public; a dedicated paper-benchmark scoring driver was not established in the inspected tree. https://github.com/apple-aiml-research/ml-fastvlm. [4].

TI-DPO (E2), ICLR 2026, Sec. 5/Table 1 [18]. The six benchmarks are MMLU, GSM8K, GPQA, HumanEval, TruthfulQA and IFEval, with higher reported scores preferred. They assess different outcomes and are not one cognitive scale. Table 1 reports averages across Llama-3.2-3B, Llama-3.1-8B and Mistral-7B-v0.3; some displayed columns do not reconcile with simple means of Tables B5–B7, and task-scoring variants are incompletely specified. We therefore use the adaptation mechanism and task scope, without importing the pooled numerical aggregate. Training/preference-evaluation code is public, but the inspected README assigns paper benchmarks to separate harnesses; their complete six-task scoring path was not established. No missing harness is inferred to exist in the inspected tree.

QSpec (E2), EMNLP 2025, Table 1 and Sec. 4. Code: https://github.com/hku-netexplo-lab/QSpec. The public demo supports throughput execution, while a matching task-fidelity table driver was not established. Main Fig. 3 redraws published exact-match scores; it adds no trials or uncertainty estimates. [19].

S9.3 Context, Execution, and Workflow Records (Table V)

StreamingLLM, ICLR 2024, Secs. 3.1/4.1/4.4 (stability) and 4.5 (efficiency), https://arxiv.org/abs/2309.17453. Efficiency experiments use a single A6000 with Llama-2-7B/13B. Attention sinks support bounded-memory streaming; they do not preserve arbitrary evicted factual detail. Code: https://github.com/mit-han-lab/streaming-llm. [7].

InfLLM, NeurIPS 2024, Appendix C.1, https://arxiv.org/html/2402.04617. LongBench and ∞Bench experiments use FP16 on A100/A800 GPUs. Retrieval of selected distant blocks differs from exact full-context attention. Code: https://github.com/thunlp/InfLLM. [8].

SGLang, NeurIPS 2024, Sec. 6, https://arxiv.org/html/2312.07104. Primary serving experiments include AWS G5/A10G (24 GB), with A100 settings separately reported. Agent workloads use replayed traces; serving improvements do not by themselves establish live-agent task success. The E1 basis is the bundled ReAct trace, Llama-2-7B commands and SGLang/vLLM drivers at revision `83525a1`, ReAct benchmark. This establishes fixed-call execution, not live planning; no independent run is claimed. [20].

EdgeShard, IEEE Internet of Things Journal, DOI 10.1109/JIOT.2024.3524255; inspected author fulltext https://arxiv.org/html/2405.14371v1. Physical resources include AGX Orin/Orin NX devices and an RTX 3090 server; the author version specifies WikiText-2, 32 input and 96 output tokens, and controlled link sweeps. A matching official artifact was not verified, so this result is E2 rather than a claim of replication. [11].

AutoGen, COLM 2024; inspected author v2 (3 October 2023), https://arxiv.org/abs/2308.08155v2. Published multi-agent outcomes depend on the task, model/API, prompts, and termination rules. Framework code is public at https://github.com/microsoft/autogen; API-backed outcomes are treated as E2, not fully repeatable on-device evidence. [21].

FastTTS (E1), ASPLOS 2026, 732–748; author v2, Sec. 6/Appendix B [22]. The downloadable evaluation artifact (Zenodo 17943373) contains accuracy code and 456 YAML benchmark configurations. The top-level run script currently enables a subset; rerunning other configurations requires explicit selection. This supports rerunning in principle, not an independent reproduction or mobile result. Precise goodput normalizes beam token length by mean beam completion time.

Agent-X (E2), MobiSys 2026, 144–157; author v1, Secs. 5.1, 5.4–5.5 [23]. TinyAgent test instances use deterministic tools; a matching implementation was not established. The evaluated pipeline does not establish unrestricted-tool safety or phone energy.

Cooperative caching (E2), TMC 25(10), 16651–16663; author v2, Secs. IV–V [5]. Inspected code includes unified evaluation, environment/training and SAC checkpoints; matching SFT/GRPO adapter weights were not present in the inspected tree. Device deadline/energy are separately unmeasured transfer questions, not the grading reason.

FunctionGemma/Mobile Actions (E2) [2, 3]. A first-party card/tuning/demo record, not archival replication evidence. Call scoring and a separate S25 Ultra CPU profile have different populations and a context-setting discrepancy (S11); they do not form one matched quality–energy result.

S9.4 Security and Additional Network-Application Sources

InjecAgent and AgentDojo define explicit attack/utility settings [9, 10]; they are not numerical risk estimates for every edge deployment. C1–C5 organize outcome types, while E1–E3 qualify publication and artifact support. A joint quality–resource comparison requires the same model/configuration, task population, timing boundary and device.

Cai et al. [24] is used only for the LLM roles documented in the author institution's abstract (institutional record); full experimental configuration was inaccessible. Liu et al. [25] documents prompt/service co-design. Its preprint and archival abstracts report different service-experience gains; neither is imported as a comparable numerical effect. These scope citations are not additional matrix measurements.

S9.5 Figure Provenance and Published Drawing Inputs

Figure 1 separates the task loop from resource adaptation. Its dashed feedback denotes offline or delayed task validation; online device telemetry does not supply an oracle for semantic correctness. Figure 2 is analytical: $N = 7 \times 10^9$, $L = 32$, $B = 1$, $H_{KV} = 8$, $d_h = 128$, and 16-bit KV state; weights use 4 or 16 bits. Its CSV is generated from the stated formula, not from measured hardware. Figure 3 uses the nine reported values in its supplied source CSV. Figures 4–7 are conceptual vector illustrations of placement, collaboration patterns, cross-layer coupling, and workflow evaluation. Their geometry carries no measured performance scale or calibrated decision threshold. All drawing scripts, vector/raster exports, and both source CSVs are public in the figure catalogue. Figure 3 is reproduced by QSpec drawing script from its source CSV; it is a literature replot, not a new experiment. Scientific figures contain no invented observations.

S10 Extended Optimization and Workflow Reporting Requirements

The main text explains the mechanisms and conditional choices; this section specifies additional fields for testing them. These are proposed reporting requirements grounded in the cited mechanisms, not new performance findings or a claim that every combination has been evaluated.

S10.1 Communication Objects and State Ownership

Whole-request delegation.
Record input/output bytes, preparation, transfer, remote queueing/execution and return/postprocessing. Name data and tool states allowed to leave the client. The main serial accounting applies only to nonoverlapping stages; pipelined execution follows the dependency graph.

Partitioned execution.
Record layer placement, activation/accumulation formats, transmitted tensors, per-node KV ownership and peak memory. Sweep the actual link and source node; EdgeShard's results depend on both [11]. Aggregate nominal RAM does not certify a feasible partition.

Cascades and step delegation.
Retain local rejections, escalation costs and failure energy. State router thresholds/calibration, merge/restart semantics and irreversible-action policy. EdgeWise author v2, Appendix G.2, uses Qwen3.5-9B, BF16 and vLLM on DGX Spark; this is not a phone configuration [26].

S10.2 Formats, Updates, and Modality State

Executable precision.
Alongside bit width, provide group size, scale/zero-point metadata, calibration samples, accumulation, dequantization placement, executed/fallback operators and backend revision. AWQ and MobileQuant target different quantized objects and kernel conditions [13, 14].

Adaptation and concurrency.
State frozen/trainable weights, adapter merge/switch policy, optimizer/update state and whether updates overlap serving. LoRA/QLoRA reduce trainable-state demands but do not remove the base model [27, 28]. Preserve a rollback checkpoint and report task regressions after an update.

Architecture and modality.
Separate total/resident weights, active expert parameters, KV heads, vision/audio encoders, image resolution and visual-token count. FastVLM's encoder/token changes affect visual prefill [4]; Gemma 3n's PLE caching and conditional modality loading affect resident state [29]. Neither a parameter label nor a text-decode rate identifies all these costs.

S10.3 Workflow, Control, and Decoding Tests

Trace replay versus live action.
Mark calls as replayed or agent-selected. Preserve external model/API, prompts, tool state and termination policy. SGLang trace efficiency and AutoGen task outcomes test different units [20, 21]. Report goal completion separately from call speed.

Controller freshness.
Budget observation, prompt construction, inference, parsing/validation and communication before the state/action deadline. For cooperative caching, keep cache-hit utility, parser failures/no-operation fallbacks and measured controller overhead separate [5]. A valid no-operation action does not guarantee the utility target.

Draft/verifier execution.
Report accepted/proposed tokens, verification/rejection rule, verifier precision, switching/synchronization overhead and shared-memory contention. QSpec evaluates a defined precision/verification relationship [19]; FastTTS evaluates branch scheduling and precise goodput [22]. Neither establishes arbitrary composition with offload or mobile kernels.

Matched composition.
Compare the baseline, each lever alone and their combination with the same tasks/scorer, output limits, failure accounting and device state. Expose shared implementation changes in ablations. This is diagnostic guidance, not an empirically established universal tuning order.

S10.4 Trajectory-Level Cost and Privacy Boundaries

EC^2-Bench author v2, Secs. 4–5/Appendix A.11, separates executable task completion, cloud-token/cost accounting and sensitivity-annotated exposure [30]. Its local FLOP approximation is not integrated device energy; its automated privacy judge is not a confidentiality guarantee. A deployment should record actual transfers and permitted data classes and separately validate any privacy or energy claim. These trajectory records complement the main matrices without adding a matched hardware experiment.

S11 Concrete Model and Agent Artifacts for Edge Workflows

These are role-matched illustrations accompanying the main application table, not an exhaustive model census or a cross-model performance ranking. Official records were inspected on 13 November 2026. The releases/examples predate the retained literature cutoff, but this artifact check is separate from the frozen arXiv/repository manifests at `82b6110`. Exact model, tokenizer/template, quantizer, runtime and export revisions must be pinned for an actual run. No checkpoint was executed for this revision.

S11.1 Role-Matched Model and Framework Catalogue

Table S7: Concrete artifact identifiers and appropriate scope. Publisher records establish component roles, not a matched cross-model ranking.

Component	Official identifier / path	Scope and required check
Qwen3 small dense text models	Qwen/Qwen3-1.7B; sibling releases 0.6B and 4B. Release record: official Qwen3 release.	Thinking/non-thinking modes and local runtime paths are documented. Pin mode/template and generated-work budget; measure target-task quality and resident KV rather than infer phone cost from model size.
Llama 3.2 compact text	meta-llama/Llama-3.2-1B-Instruct; 3B-Instruct sibling.	Publisher describes text retrieval/dialogue and summarization roles in constrained settings. Task language, context and application safeguards remain deployment-specific.
FunctionGemma	google/functiongemma-270m-it; model card: official FunctionGemma card.	Specialized function calling, not direct general dialogue. Application code must parse, authorize and execute a call. Domain tuning and schema-specific evaluation are separate from runtime availability.
Gemma 3n	google/gemma-3n-E2B-it; E4B-it sibling. Overview: official Gemma 3n overview.	Text, visual and audio input; E2B/E4B are effective-size labels. Per-layer embedding (PLE) caching and conditional modality loading alter resident state, so the label is not total checkpoint size or application RAM.
SmolLM2	HuggingFaceTB/SmolLM2-1.7B-Instruct; 135M/360M siblings.	Publisher supplies CPU/GPU inference, rewriting and summarization examples. Check supported language, prompt and long-context limits; low parameter count is not a correctness guarantee.
FastVLM	https://github.com/apple-aiml-research/ml-fastvlm; 0.5B/1.5B/7B checkpoints, `model_export` and `app`.	Official Apple-device demo/export artifacts are distinct from the paper's M1 measurements. Pin vision encoder, image resolution and model/quantization stage before device profiling.
Octopus-v2	NexaAI/Octopus-v2; paper arXiv:2404.01744.	Early Android-API functional-token example. The card's 2B-class naming is not used to infer loaded RAM. Its benchmark setting and tool assumptions cannot establish generic phone-agent success.
AutoGen	https://github.com/microsoft/autogen; COLM 2024 paper; inspected author v2 (arXiv:2308.08155v2). A matching historical code revision was not established; the live repository is a separate artifact.	Orchestration framework, not an LLM or mobile measurement. Freeze the model backend, tools, prompts and termination policy; model/API changes affect task outcomes.

S11.2 Documented Application and Device-Profile Boundaries

Documented application versus hypothetical composition. Google's Mobile Actions in AI Edge Gallery uses FunctionGemma to select OS functions, while Tiny Garden illustrates app-specific action schemas. The official account is dated 26 February 2026 and documents Android/iOS demonstration paths: https://developers.googleblog.com/on-device-function-calling-in-google-ai-edge-gallery/. These are concrete function-use examples; broad autonomy, safety and task energy need their own evaluations [3].

First-party device-profile boundary. The publisher reports 58% versus 85% call accuracy after Mobile Actions tuning. Its separate S25 Ultra CPU profile uses LiteRT/XNNPACK, four threads and dynamic INT8; the table lists TTFT 0.3 s and peak RSS 551 MB. Table context is 1024, whereas the prose states 512 prefill plus 32 decode tokens. No uncertainty or task energy is supplied. We retain these settings without reconciling them or equating score and device populations. The main matrix labels this record E2 [2].

Model, device and workflow are distinct units. Measure actual loaded weights, modality encoders, KV, workspace and application reservation. Report cold/warm loading, context/output lengths, sustained thermal state and any radio/host energy. A mobile CPU, phone GPU/NPU, embedded GPU and shared edge server require different executable formats and concurrency policies. Runtime support is a compatibility prerequisite, not evidence of a particular end-to-end rate. Specify tool connectivity and safe retry conditions.

S12 Extended Private-Assistant Decision and Acceptance Contract

This extends the main Worked Example with a prospective trace. It supplies a complete use of the capability index, evidence matrices, placement check and evaluation boundary; it adds no experimental observation. StreamingLLM/InfLLM motivate the distinct context choices, and FunctionGemma documents a function-calling component [7, 8, 2]. The goal is to retrieve a constraint from a user's private notes and create one reminder after confirmation. The task requires the correct source fact, valid tool arguments, an authorized single action and a declared terminal state.

S12.1 Workflow and Acceptance Stages

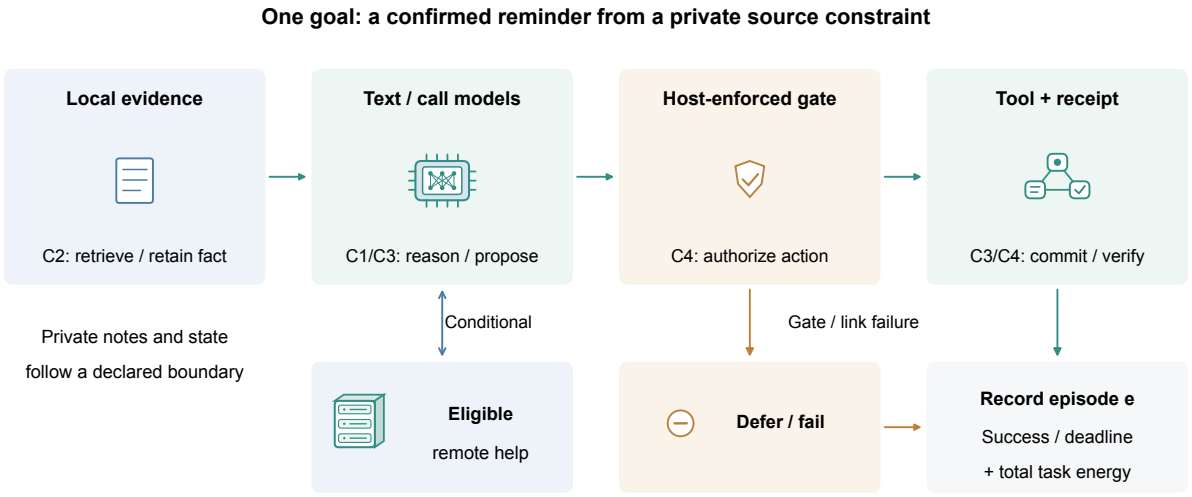


Figure S3: A prospective private-assistant workflow. Local evidence and a compact text model feed a schema/permission/confirmation gate before a tool action. The host checks the result and records the whole episode. Delegation is conditional on data admissibility and feasibility; a failed gate returns a declared defer/fail outcome. These are proposed component roles, not measured gains or a claim that the combined system was executed.

Table S8: From task capability to an inspectable deployment choice. The supporting sources justify individual mechanisms, not the unmeasured combined pipeline.

Stage	Choice and evidence boundary	Record and acceptance
Declare goal	C2 requires the earlier fact; C3/C4 require one correct, permitted reminder. A fluent response is not completion.	Episode ID, source-fact criterion, allowed action, confirmation rule, deadline D, device memory and energy boundaries.
Diagnose state	Profile weights, KV and workspace separately. Streaming continuity does not imply access to evicted facts; selected retrieval is a different C2 candidate.	Actual context lengths, retrieved source IDs/freshness, critical-fact retention, resident memory and lookup cost.
Select text model	A compact Qwen3/Llama component is a candidate. Fix template, precision and reasoning/output cap; neither a card nor a next-token proxy validates this task.	Checkpoint/runtime revisions, task-correct evidence use, rejected/extra generations and precision-induced failures.
Generate action	A specialized function caller can target the bounded API; a second model adds loading and state costs.	Function schema, all arguments, schema-validation failures, model/tool call IDs and parent links.
Authorize and commit	Host policy, not model confidence, enforces permissions and confirmation. Use an idempotency key or equivalent result check before retrying.	Authorization result, user confirmation, action receipt, duplicate/unsafe actions, rollback or recovery state.

Continued on the next page

Stage	Choice and evidence boundary	Record and acceptance
Place/degrade	Delegate only eligible inputs when transfer, queueing, execution and return fit the budget. If no path is adequate, declare defer/failure.	Bytes/data classes crossing the boundary, link/queue state, fallback path, timeout and terminal reason.
Evaluate	Keep every episode in A; count in S only verified correct single-action completion. Measure success by D separately.	S/A, success-before-deadline rate, request/task tails, retries, mean attempt energy and amortized success energy (S7), declared component power window.

S12.2 Trace Linkage, Recovery, and Matched Tests

Trace linkage and failure interpretation. An illustrative dependency is model request `r1`, retrieval/tool call `t1`, a follow-up request `r2`, and the authorized commit `t2`, all linked to one episode. Tool re-execution after a timeout can duplicate a side effect even when each request returns valid JSON. Preserve the original action receipt and distinguish an unconfirmed action, a committed action with a missing response, and an explicitly rejected action. Any retry remains inside the same task episode until the declared outcome; a new user restart is a new episode. If commitment cannot be established, record an indeterminate outcome, avoid blind replay, and exclude it from S while retaining it in A. The request helper in S7 does not infer these goal-level states.

Matched tests required for the composition. Compare the feasible baseline with one intervention on context, precision or depth and their combination using the same task instances, scorer, output caps, permissions and device state. Vary stale/missing evidence, link outage, tool timeout and sustained-session temperature. Attribute all local/remote model, tool and radio costs under declared windows; avoid double-counting overlapping shared-device energy.

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